

SUBORNO DEB BAPPON

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EDUCATION

University of Saskatchewan

Sep. 2023 – Present

Master of Science in Computer Science – Grade: **89.5%**

Saskatoon, SK

- Coursework: Software Maintenance & Evolution, Empirical Software Engineering, Advanced Deep Learning, Human-Centric Software Renovation

Chittagong University of Engineering & Technology

Feb. 2017 – Sep. 2022

Bachelor of Science in Computer Science & Engineering – CGPA: **3.76/4.00 (Merit: 3rd among 128)**

Chattogram, Bangladesh

- Coursework: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Database Management Systems, Operating Systems

SKILLS

Programming Languages: Python, C, C++, JavaScript, HTML, CSS

Frameworks & Libraries: Flask, Django, Gradio, LangChain, OpenAI Agents SDK, CrewAI

Machine Learning & Data Analytics: scikit-learn, PyTorch, TensorFlow, NumPy, Pandas, Matplotlib, Seaborn, SQL, Power BI

Tools & Platforms: Git, GitHub, Azure, VS Code, JupyterLab, Linux, Windows, macOS

EXPERIENCE

Extreme Solutions Ltd.

Jan. 2022 – Aug. 2022

Junior Data Scientist

Chattogram, Bangladesh

- Cleaned, analyzed, and visualized datasets using *Python (Pandas, NumPy, Matplotlib)* to uncover actionable insights that improved data-driven decision-making by 20%.
- Developed and trained predictive models with *scikit-learn* to forecast customer trends, achieving 85% accuracy and supporting marketing optimization efforts.
- Automated data preprocessing and reporting workflows using *SQL* and *Python scripts*, reducing manual data handling time by 30%.

University of Saskatchewan

Sep. 2023 – Present

Research Assistant — Software Research Lab

Saskatoon, SK

- Built scalable *Python* pipelines for large-scale *Stack Overflow* and *GitHub* data, improving data processing efficiency and enabling deeper developer behavior insights.
- Developed LLM-powered tools that enhanced code comment quality by 30% and achieved 84.5% adoption among developers for *Stack Overflow* answer refinement.
- Reduced software prototyping time by 60% and documentation workload by 80% via multi-agent AI systems for automated software design and documentation.

University of Saskatchewan

Sep. 2023 – Present

Teaching Assistant — Dept. of Computer Science

Saskatoon, SK

- Guided 30+ students through data-driven programming projects, improving project completion rates by 20% through hands-on mentorship and code review sessions.
- Designed and implemented structured grading rubrics that identified key learning gaps, improving average final exam performance by 15%.
- Mentored students on software documentation, version control (Git), and code quality standards, resulting in a 35% increase in industry-ready project submissions.

Eastern University

Feb. 2023 – Jul. 2023

Lecturer — Computer Science & Engineering

Dhaka, Bangladesh

- Delivered lectures and labs on Python, databases, and machine learning, integrating real-world projects that boosted student lab performance by 22%.
- Mentored undergraduates in software development best practices, improving project completion rates by 30%.
- Modernized course content to align 80% of curriculum with industry topics like data analytics, machine learning and software engineering, enhancing internship readiness.

PROJECTS

Neora: Multi-Agent System for Rapid Software Prototyping – *Python, CrewAI, OpenAI/Gemini API, Gradio, Docker*

[GitHub]

- Developed an autonomous multi-agent system that converts plain-text requirements into Python modules and UI prototypes, streamlining the MVP development process.
- Engineered modular YAML pipelines enabling adaptive role-specific reasoning across LLMs such as GPT-5 and Gemini 2.5 Flash.

Repolish: Automated README File Generator – *Python, CrewAI, OpenAI/Gemini API, Gradio, CSS*

[GitHub]

- Built an AI-powered tool to parse GitHub repositories and auto-generate professional README files, optimizing documentation process.
- Integrated Gradio-based LLM interface and automated PR workflows, reducing release cycles from hours to minutes.

DEBug.me: Personal Agentic AI Assistant – *Python, RAG, OpenAI Embeddings, Gemini API, Chroma, Gradio*

[Hugging Face]

- Built a RAG-powered AI assistant using Chroma vector search and multi-API integration for accurate, context-grounded responses.
- Deployed a Gradio-based web app with real-time chat, conversation memory, and NTFY-integrated user notifications.

Farmona: Smart Crop Prediction & Agronomic Analysis – *Python, Scikit-learn, Pandas, NumPy, Streamlit*

[GitHub]

- Built an ML pipeline with feature engineering and GridSearchCV optimization, achieving 99% accuracy in crop prediction.
- Deployed an interactive Streamlit app for real-time agronomic insights with reproducible deployment artifacts.

SELECTED PUBLICATIONS

1. **Suborno Deb Bappon**, Saikat Mondal, and Banani Roy. “AUTOGENICS: Automated Generation of Context-Aware Inline Comments for Code Snippets on Programming Q&A Sites Using LLM.” In *2024 IEEE International Conference on Source Code Analysis and Manipulation (SCAM)* (pp. 24–35). IEEE.

2. Saikat Mondal, **Suborno Deb Bappon**, and Chanchal K. Roy. “Enhancing User Interaction in ChatGPT: Characterizing and Consolidating Multiple Prompts for Issue Resolution.” In *2024 IEEE/ACM 21st International Conference on Mining Software Repositories (MSR)* (pp. 222–226). IEEE.

CERTIFICATIONS

Machine Learning with Python (IBM), Neural Networks & Deep Learning (DeepLearning.AI), Natural Language Processing with TensorFlow (DeepLearning.AI), Problem Solving (Intermediate) (HackerRank)

HONORS & AWARDS

Best Research Award, CMPT 854: Human-Centric Software Renovation, University of Saskatchewan (Spring 2024)

75th Anniversary Recruitment Scholarship, CGPS, University of Saskatchewan (2023 – 2025)

Dean's Award, Faculty of Electrical & Computer Engineering, Chittagong University of Engineering & Technology (Sep. 2022)

EXTRACURRICULAR ACTIVITIES

Vice President Internal, Computer Science Graduate Council, University of Saskatchewan (Sep. 2024 – Oct. 2025)

Mentor, Eastern University Computing Club (May 2023 – Jul. 2023)

REFERENCES

Dr. Chanchal K. Roy, Professor, Department of Computer Science, University of Saskatchewan — chanchal.roy@usask.ca, (306) 715-0600